Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Harm, Deborah Ph.D.		
Project Title:	Virtual Reality Training: Cybersickness and Effects on Sensorimotor Functions		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	10/01/1999	**End Date:**	03/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	N/A		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Kennedy, Robert (RSK Assessments) Bloomberg, Jacob (NASA JOhnson Space Center) Parker, Donald (University of Washington) Reschke, Millard (NASA Johnson Space Center) Loftin, Bowen (Old Dominion University)		
Grant/Contract No.:	None		
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Performance Goal Text:			

Task Description:	Though often brilliantly engineered, two critical and unresolved human factors issues in VRs are: 1) potential "cybersickness", a form of space motion sickness, which is experienced in virtual worlds, and 2) maladaptive sensorimotor responses following virtual environment (VE) training. Although, numerous studies show motion sickness and disturbances in sensorimotor functions following flight simulator training, during and following space flight, similar research concerning the aftereffects of VR training is only beginning to appear in literature. The plasticity of the human central nervous system makes it likely that users will eventually adapt to any unpleasant perceptual experiences in a virtual environment. However, the most critical problem for training applications is that behaviors learned in the VE may not be similar to the responses required in the real environment, potentially resulting the acquisition of inappropriate perceptual-motor behaviors. This study proposes to evaluate and compare responses to two types of VR delivery systems (head-mounted display [HMD] and a dome-projection system), two exposure duration periods (30 minand 60 min), and repeated exposures (three sessions). Specific responses that we will examine include cybersickness severity and symptom patterns, and several sensorimotor functions (eye-head and eye-hand coordination, postural equilibrium).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall goal of the proposed research is to examine the extent to which "immersion" in VR systems produces motion sickness and disrupts perceptual and sensorimotor functions that may ultimately result in negative impacts on performance of operational activities.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 11/17/2010)